



# Welcome to CSC 365

Internet Programming

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What font type and size does `<h1>Introduction</h1>` generate?

Answer: Some default from the browser (HTML tells what browser how)

Early HTML - Override defaults with attributes

```
<table border="2" bordercolor="black">
```

Style sheets were added to address this:

Specify style to use rather than browser default

Not have to code styling on every element

## **Key concept: Separate style from content**

Content (what to display) is in HTML files

Formatting information (how to display it) is in separate style sheets (.css files).

Use an element attribute named `class` to link (e.g. `<span class="test">`)

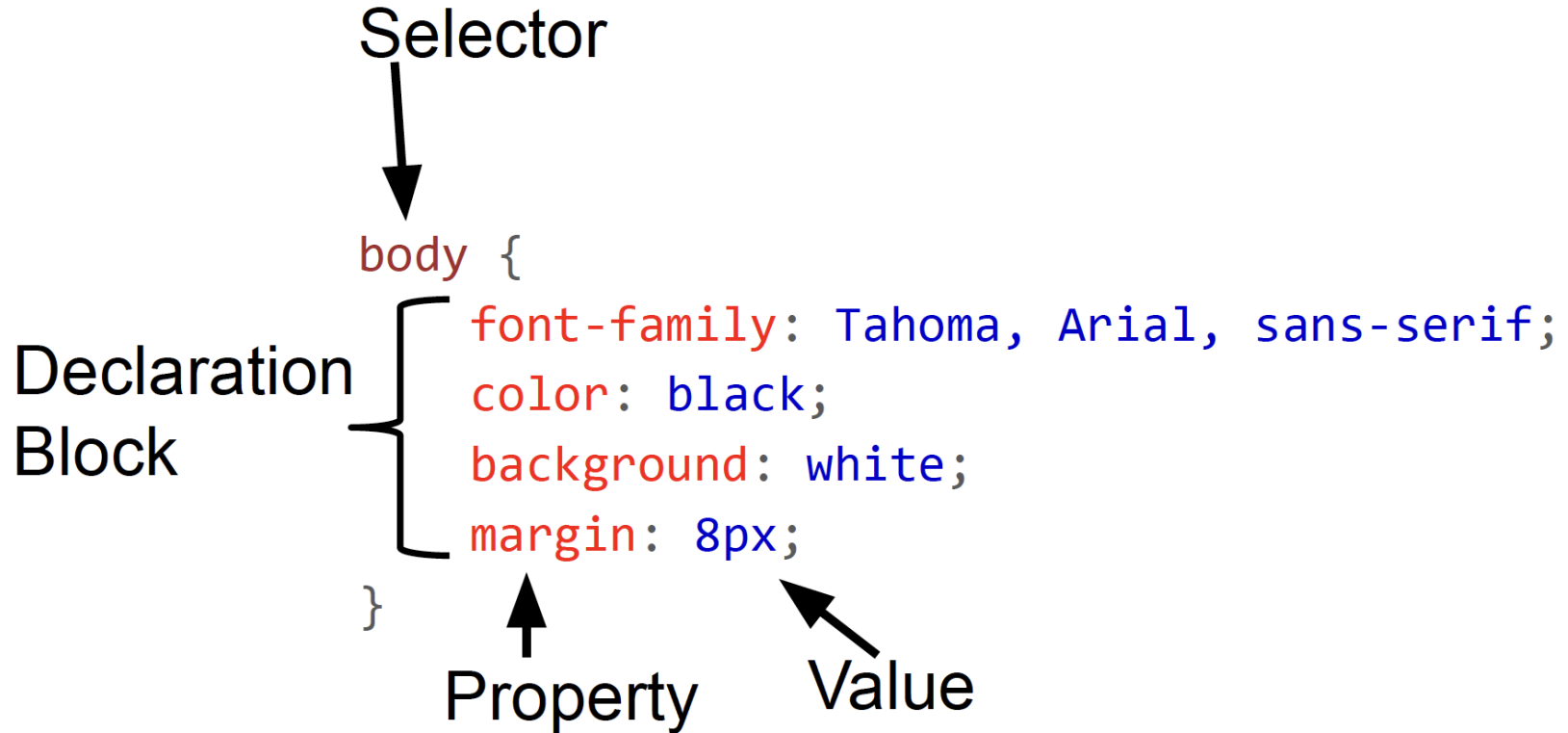
Result: define style information once, use in many places

Consider can you make all the text in the app slightly bigger?

Or purple is our new company color.

**DRY principle: Don't Repeat Yourself**

# Style sheet contain one or more CSS Rules



# CSS Selector

| CSS Selector    | CSS                                              | HTML                                             |
|-----------------|--------------------------------------------------|--------------------------------------------------|
| Tag name        | <pre>h1 {<br/>  color: red;<br/>}</pre>          | <pre>&lt;h1&gt;Today's Specials&lt;/h1&gt;</pre> |
| Class attribute | <pre>.large {<br/>  font-size: 16pt;<br/>}</pre> | <pre>&lt;p class="large"&gt;...</pre>            |
| Tag and Class   | <pre>p.large {...}</pre>                         | <pre>&lt;p class="large"&gt;...</pre>            |
| Element id      | <pre>#p20 {<br/>  font-weight: bold;<br/>}</pre> | <pre>&lt;p id="p20"&gt;...</pre>                 |

# CSS Pseudo Selectors

hover - Apply rule when mouse is over element (e.g. tooltip):

```
p:hover, a:hover {  
    background-color: yellow;  
}
```

a:link, a:visited - Apply rule when link has been visited or not visited (link)

```
a:visited {  
    color: green;  
}  
a:link {  
    color: blue;  
}
```

# CSS Properties

Control many style properties of an element:

- Coloring
- Size
- Position
- Visibility
- Many more: (e.g. `p: { text-decoration: line-through; }`)
- Also used in animation:

[https://www.w3schools.com/css/css3\\_animations.asp](https://www.w3schools.com/css/css3_animations.asp)

# Color - Properties: color & background\_color

Must ultimately turn into red, green, and blue intensities between 0 and 255:

Predefined names: red, blue, green, white, etc. (140 standard names)

8-bit hexadecimal numbers for red, green, blue: #ff0000



0-255 decimal intensities: rgb(255,255,0)



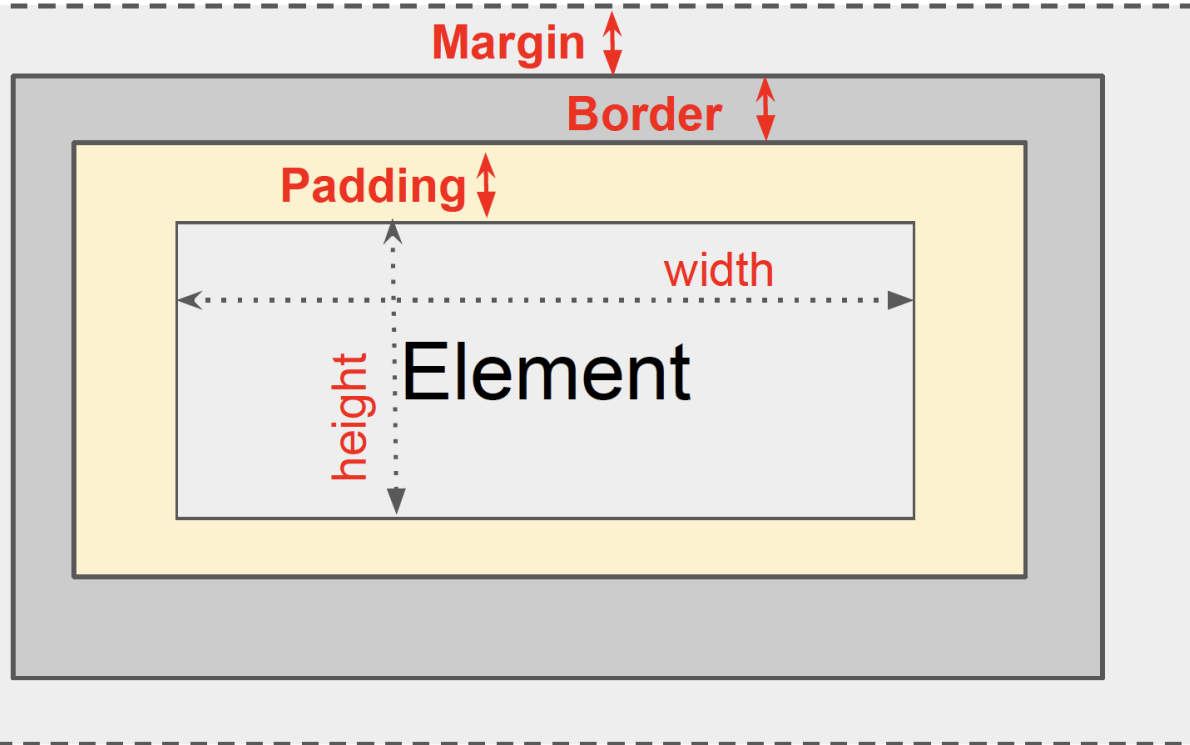
Percentage intensities: rgb(80%,80%,100%)



h1: { color: red; }



# CSS Box Model



**Total element width =**  
width +  
left padding +  
right padding +  
left border +  
right border +  
left margin +  
right margin

Margin & Padding  
Transparent

# CSS distance units

| Absolute |                                              |
|----------|----------------------------------------------|
| 2px      | pixels                                       |
| 1mm      | millimeters                                  |
| 2cm      | centimeters                                  |
| 0.2in    | inches                                       |
| 3pt      | printer point 1/72 inch                      |
| Relative |                                              |
| 2em      | 2 times the element's current font size      |
| 3rem     | 3 times the root element's current font size |

# Size Properties - Element, pad, margin, border

|                                   |                     |
|-----------------------------------|---------------------|
| width - Override element defaults | border-bottom-color |
| height                            | border-bottom-style |
| padding-top                       | border-bottom-width |
| padding-right                     | border-left-color   |
| padding-bottom                    | border-left-style   |
| padding-left                      | border-left-width   |
| margin-top                        | border-right-color  |
| margin-right                      | border-right-style  |
| margin-bottom                     | border-right-width  |
| margin-left                       | etc.                |

```
p: { border: 5px solid red; }
```

# position property

**position:** static;

(default) - Position in document flow

**position:** relative;

Position relative to default position via top, right, bottom, and left properties

**position:** fixed;

Position to a fixed location on the screen via top, right, bottom, and left properties

**position:** absolute;

Position relative to ancestor absolute element via top, right, bottom, and left properties

Fixed position (0,0) is top left corner

## Some more common properties

**background-image**: image for element's background

**background-repeat**: should background image be displayed  
in a repeating pattern (versus once only)

**font, font-family, font-size, font-weight, font-style**: font  
information for text

**text-align, vertical-align**: center, left, right

**cursor**: Set the cursor when over element (e.g. help)

# Element visibility control properties

**display:** **none**; - Element is not displayed and takes no space in layout.

**display:** **inline**; - Element is treated as an inline element.

**display:** **block**; - Element is treated as a block element.

**display:** **flex**; - Element is treated as a flex container.

**display:** **grid**; - Element is treated as a grid container.

**visibility:** **hidden**; - Element is hidden but space still allocated.

**visibility:** **visible**; - Element is normally displayed

# Flexbox and Grid layout

- `display: flex;` (Flexbox)
- `display: grid;` (Grid) newer layout method
  - Items flex to fill additional space and shrink to fit into smaller spaces.
  - Useful for web app layout:
    - Divide up the available space equally among a bunch of elements
    - Align of different sizes easily
    - Key to handling different window and display sizes
- Flexbox - Layout one dimension (row or column) of elements
- Grid - Layout in two dimensions (rows and columns) of elements

# Some other CSS issues

- Inheritance

- Some properties (e.g. font-size) are inherited from parent elements
- Others (border, background) are not inherited.

- Multiple rule matches

- General idea: most specific rule wins

`<span>Text1</span>`

`<span class="test">Text2</span>`

`span.test { color: green }`

`span { color: red }`



# Next Class



Week 2:

Topic:

- CSS